

Lecture 9: Introduction to Cognitive Robotics

COMP3018/COMP5018: Human-Robot Interaction

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Introduction

- Today's Topics

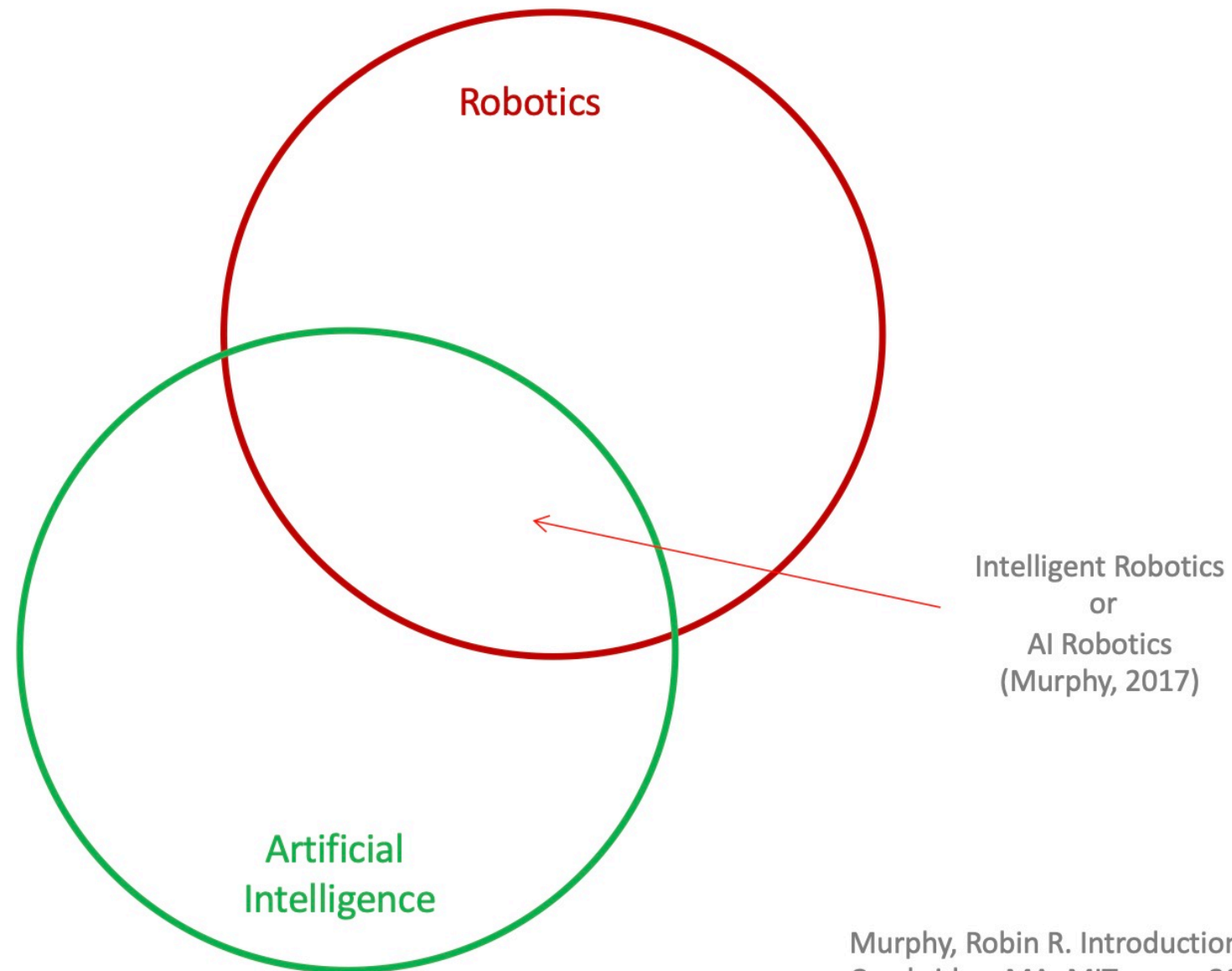
- 1- Introduction to cognitive robotics

- Session Learning Outcomes

By the end of today's lecture you will be able to:

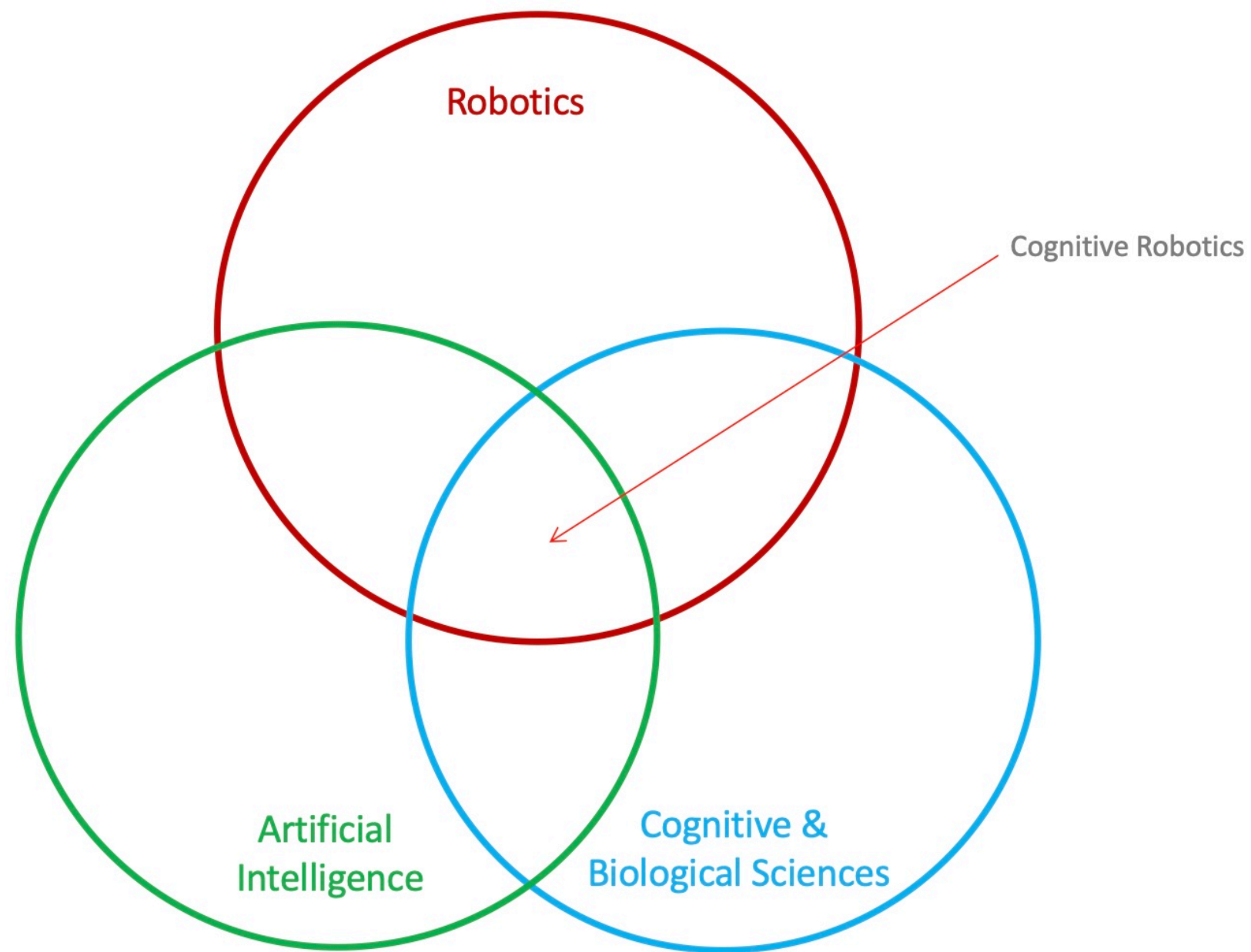
- 1- Describe different cognitive functionalities that a robot can have

What is Cognitive Robotics?



Murphy, Robin R. Introduction to AI Robotics.
Cambridge, MA: MIT press, 2019.

What is Cognitive Robotics?



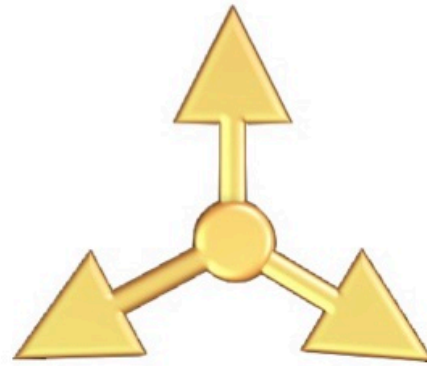
A Definition of Cognitive Robotics

"Cognitive Robotics is the field that combines insights and methods from robotics, AI and cognitive and biological sciences to design an integrated cognitive system combining the sensorimotor behavior and higher-level functions and social capabilities of an intelligent robot"

A. Cangelosi and M. Asada, Cognitive Robotics, Chapter 1, MIT Press, in press.

Cognitive Robotics Emphasizes ...

Bio-inspired
human-like and animal-like
behaviour and intelligence

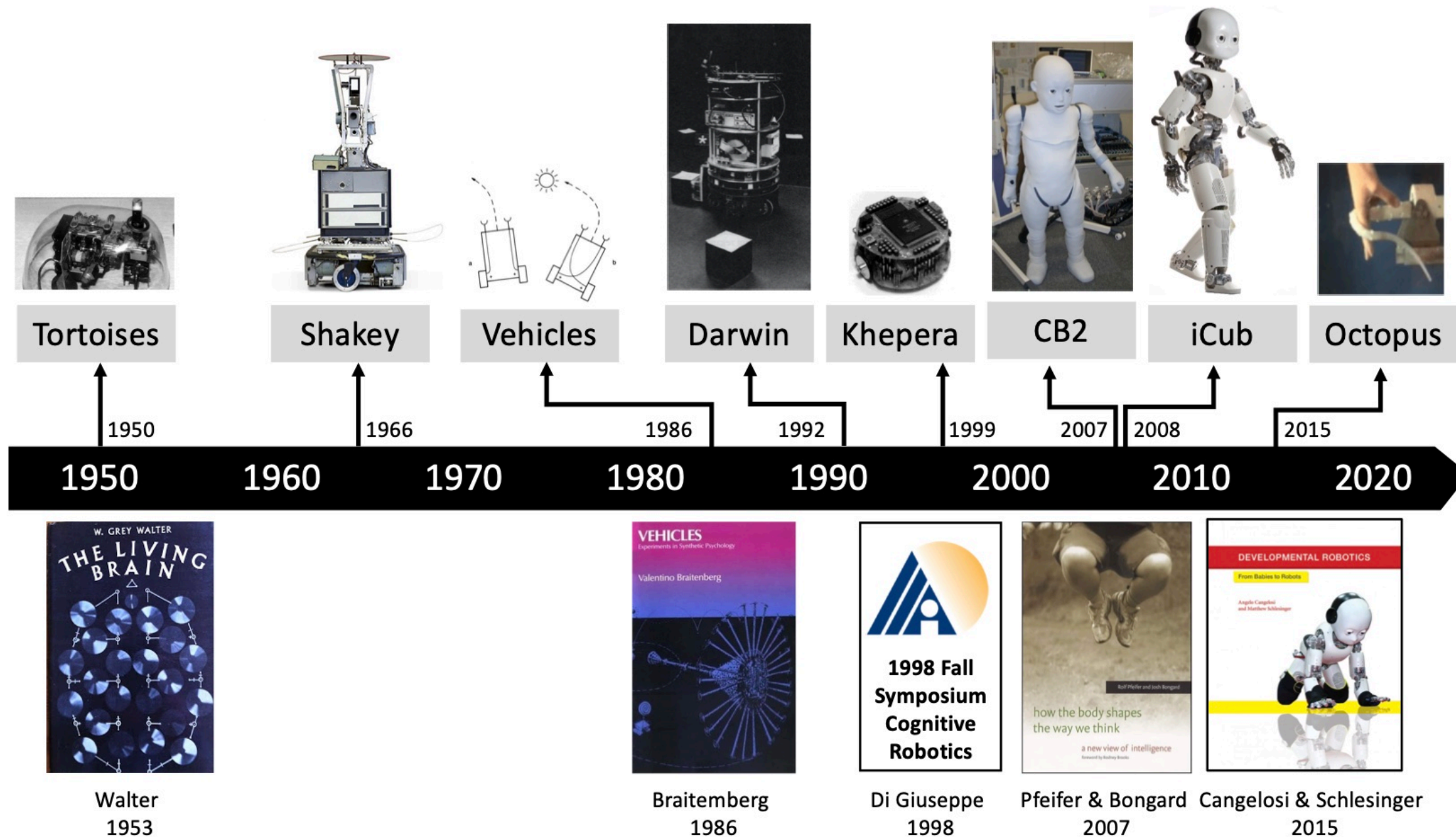


System-level integration
of a range of cognitive abilities:

- Sensorimotor skills
- Knowledge representation & reasoning
- Social interaction

Interdisciplinary approach,
including cognitive (neuro)science,
cognitive psychology, and biology

Development of Cognitive Robotics



A. Cangelosi and M. Asada, Cognitive Robotics, Chapter 1, MIT Press, in press.

Still not Clear ...!

What is Cognitive Robotics?

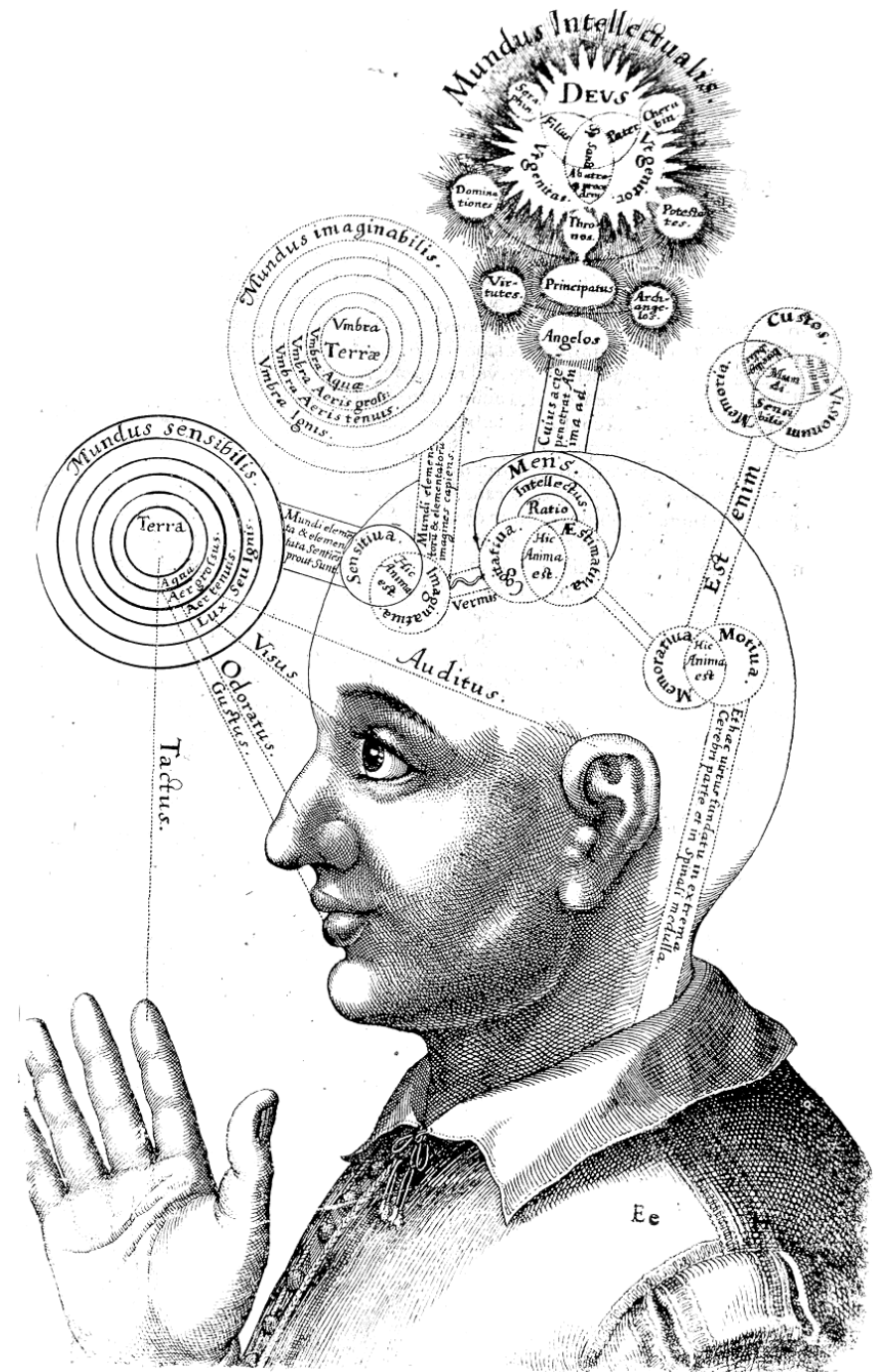
What is Cognitive Robotics?

What is cognition?

Why is cognition useful in robotics?

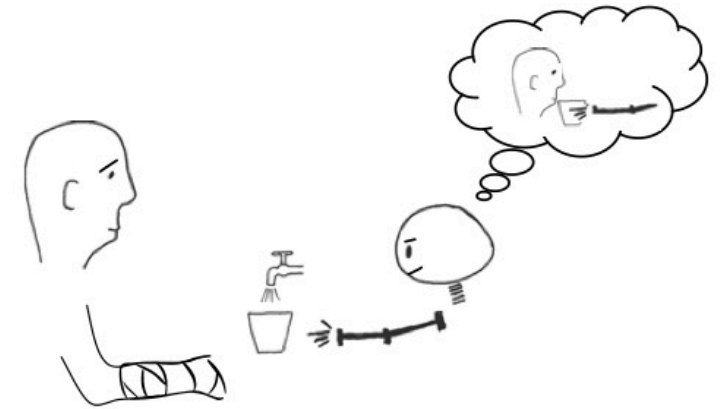
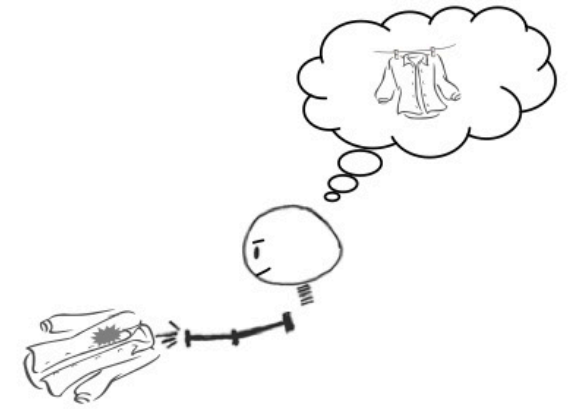
We answer
this question
first

How do cognitive robots work?



Why is Cognition Useful in Robotics?

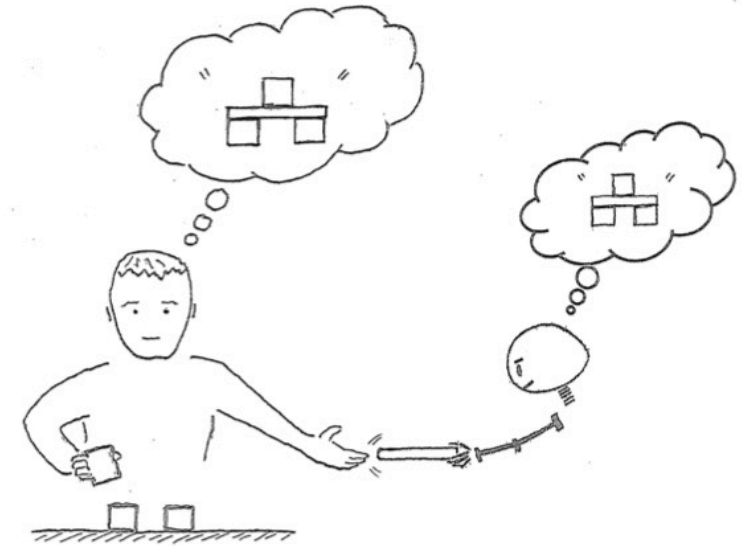
1. It allows robots to work **autonomously** in **everyday environments**
2. It enables **effective interaction** with **humans**



Why is Cognition Useful in Robotics?

- A cognitive robot is capable of **flexible context-sensitive goal-directed action**
- The robot **anticipates**
 - the need to act
 - the outcome of the action
- The action itself is guided by **prospection**
- It can also **adapt** to changing circumstances
 - Adjusting existing action policies
 - Creating new action policies when required
- It does this by dynamically recruiting several **core cognitive abilities**:
perception, attention, action selection, memory, learning, reasoning, metacognition, prospection

The robot's need to act and a human's need to act



What is Cognitive Robotics?

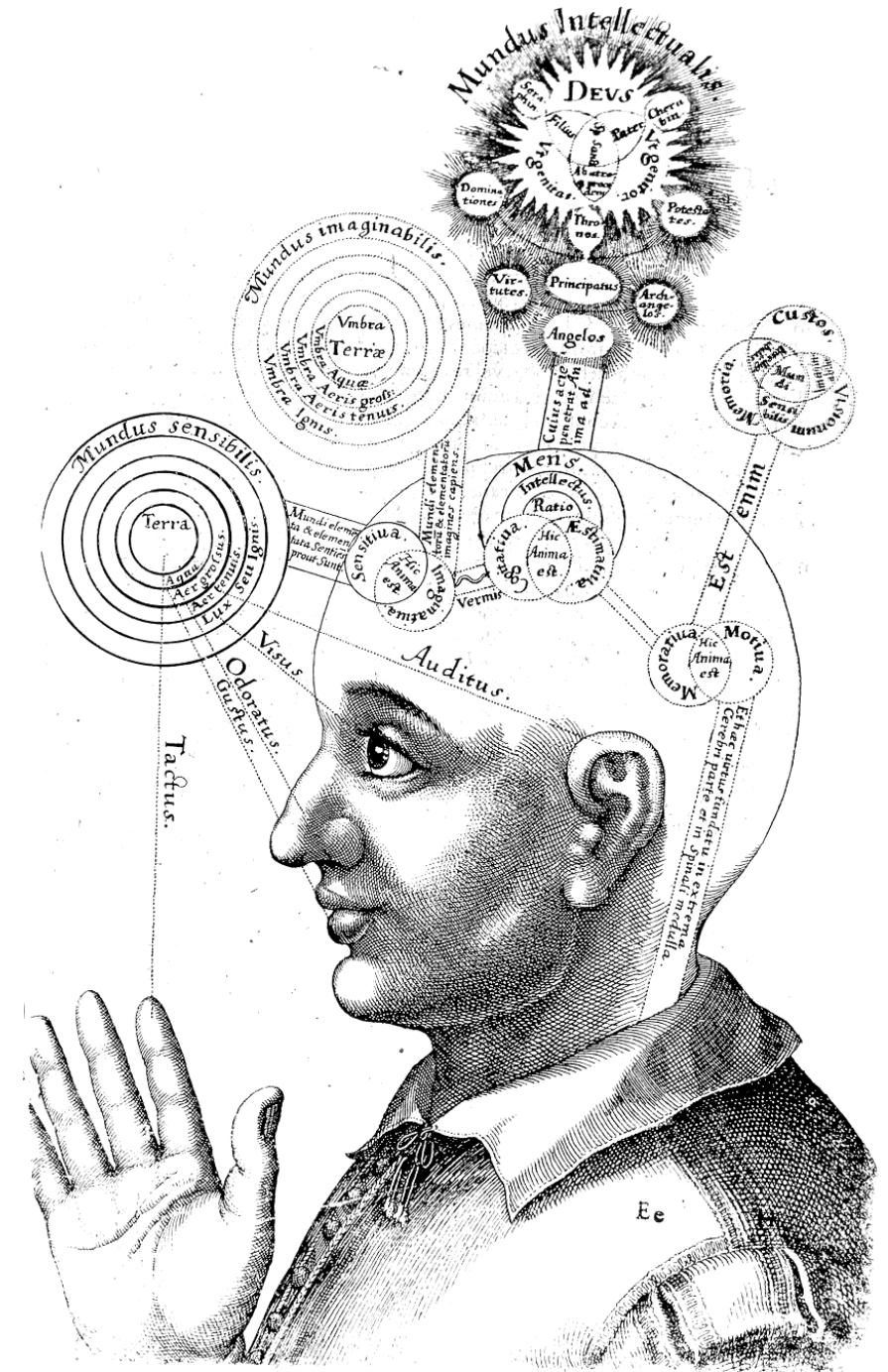
What is cognition?



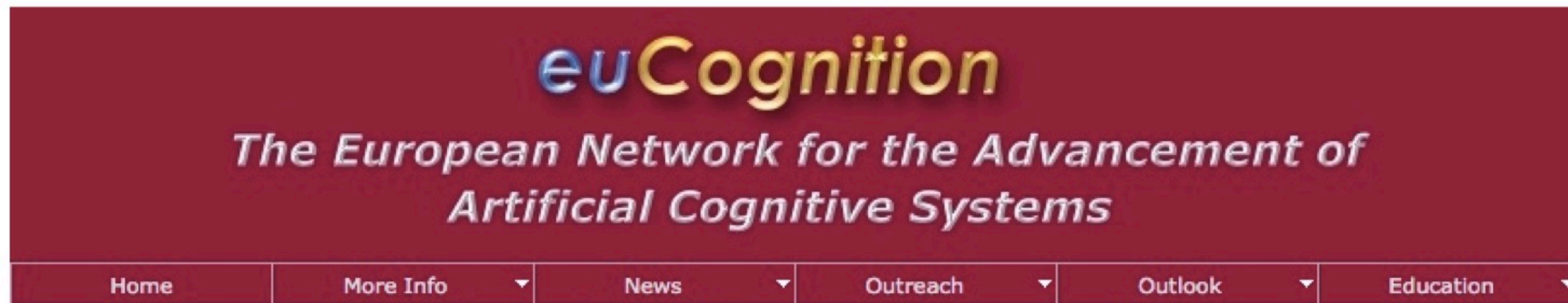
Now we answer
this question

Why is cognition useful in robotics?

How do cognitive robots work?



What is Cognition?



Definitions of Cognition & Cognitive Systems

The following definitions were contributed by members of euCognition in response to a [questionnaire](#). If you haven't completed the questionnaire, please consider doing so.

The definitions are listed in the order in which they were submitted.

Cognition is the ability to relate perception and action in a meaningful way determined by experience, learning and memory.

Mike Denham

A cognitive system possesses the ability of self-reflection (or at least self-awareness).

Horst Bischof

Cognition is gaining knowledge through the senses.

Majid Mermehdi

Cognition is the ability to ground perceptions in concepts together with the ability to manipulate concepts in order to proceed toward goals.

Christian Bauckhage

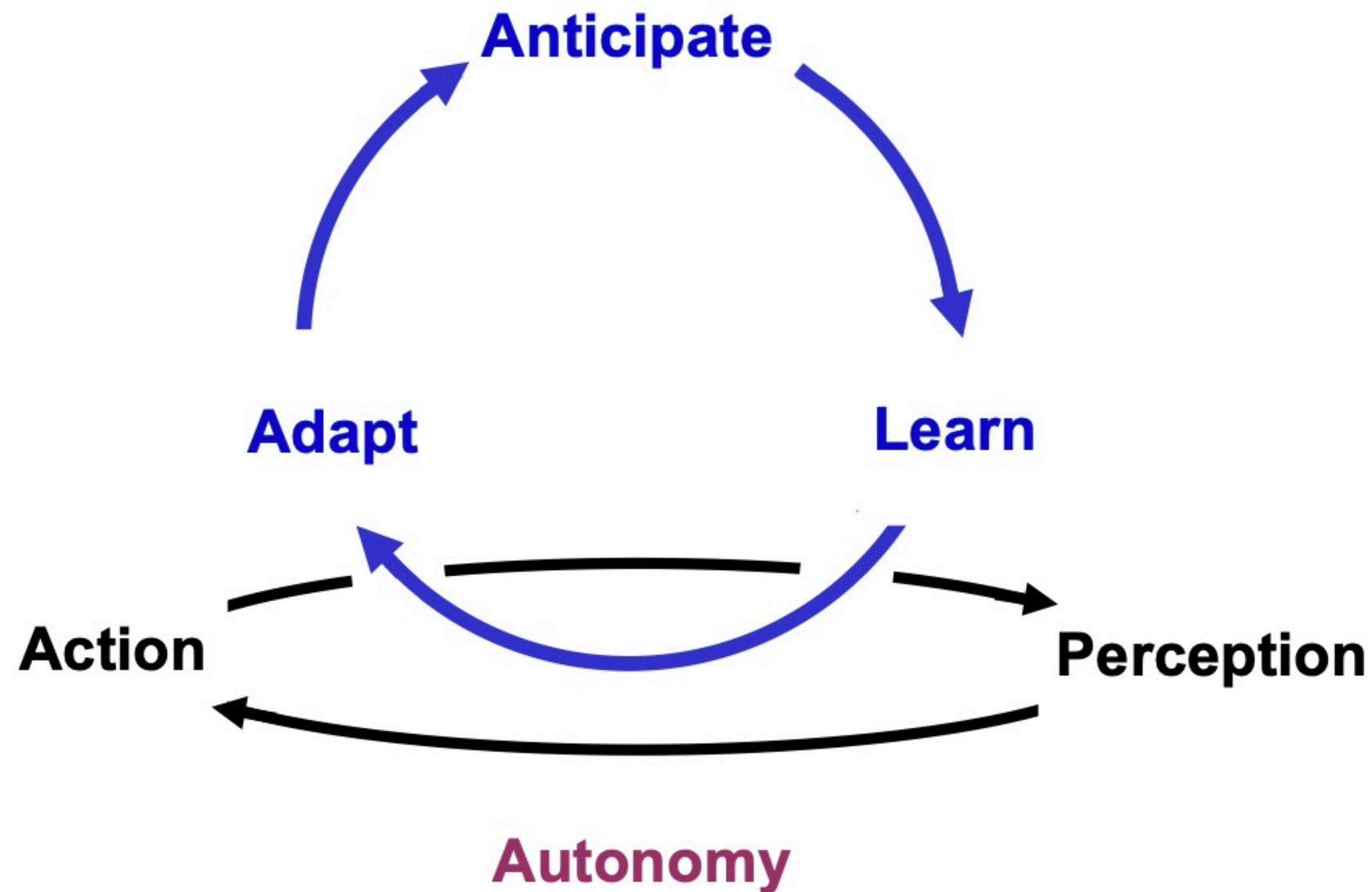
An artificial cognitive system is a system that is able to perceive its surrounding environment with multiple sensors, merge this information, reason about it, learn from it and interact with the outside world.

Barbara Caputo

42 definitions



What is Cognition?



D. Vernon, Artificial Cognitive Systems – A Primer, MIT Press, 2014

What is Cognition?

Core abilities of a cognitive system

Perception

Attention

Action selection

Memory

Learning

Reasoning

Meta-reasoning

... and **prospection**

What is Cognition?

... and **prospection**



Anticipation

Prediction

Intention

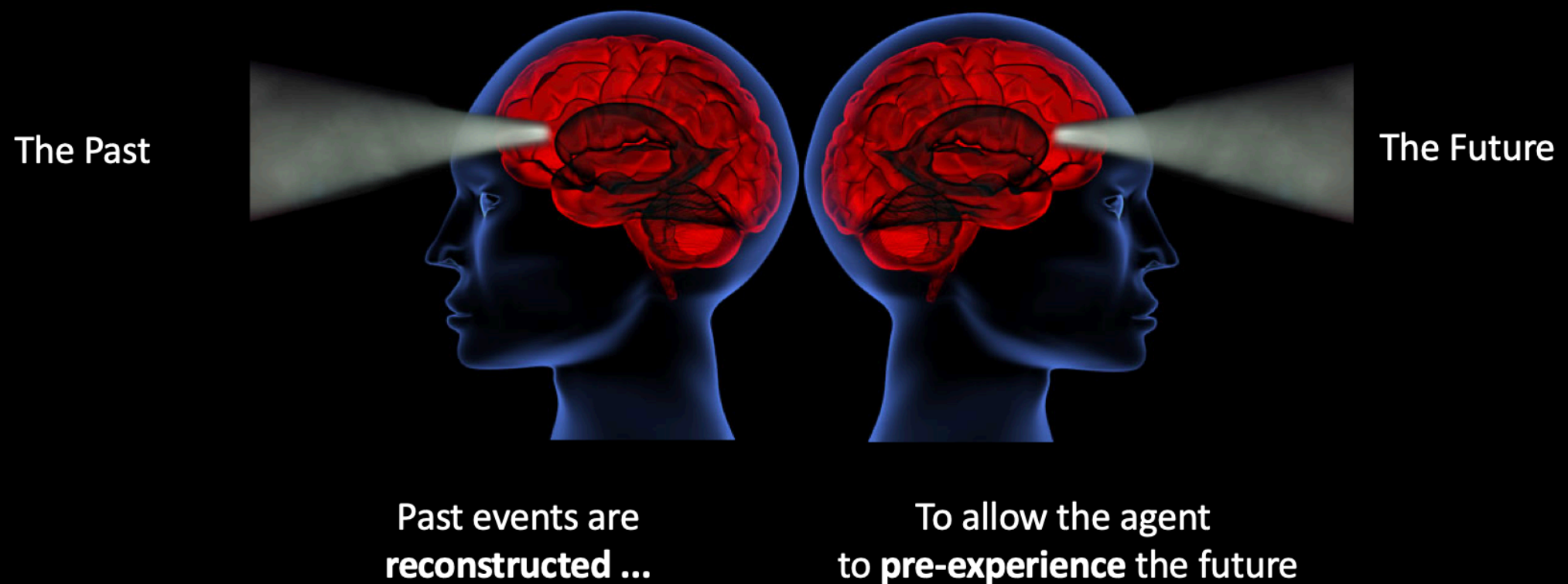
Planning

Simulation

Episodic future thinking

What is Cognition?

Episodic Future Thinking



C. M. Atance and D. K. O'Neill, "Episodic future thinking," Trends in Cognitive Sciences, vol. 5, no. 12, pp. 533–539, 2001.

D. L. Schacter and D. R. Addis, "The cognitive neuroscience of constructive memory: Remembering the past and imagining the future," Philosophical Transactions of the Royal Society B, vol. 362, pp. 773–786, 2007.

What is Cognition?

A More Detailed Definition of Cognitive Robotics

"The word cognition derives from the Latin verb cognosco, a composition of con (meaning related to) and gnosco (to know). Cognitive robotics, then, is the branch of robotics where **knowledge** plays a central role in supporting **action selection**, **execution**, and **understanding**.

It focuses on designing and building robots that have the ability to **learn from experience** and **from others**, commit relevant knowledge and skills to **memory**, retrieve them as the **context** requires, and **flexibly** use this knowledge to select appropriate actions in the pursuit of their **goals**, while **anticipating** the outcome of those actions when doing so.

Cognitive robots can use their knowledge to **reason about their actions** and the actions of those with whom they are **interacting**, and thereby **modify their behavior** to improve their overall long-term effectiveness.

In short, **cognitive robots are capable of flexible, context-sensitive action, knowing what they are doing and why they are doing it."**

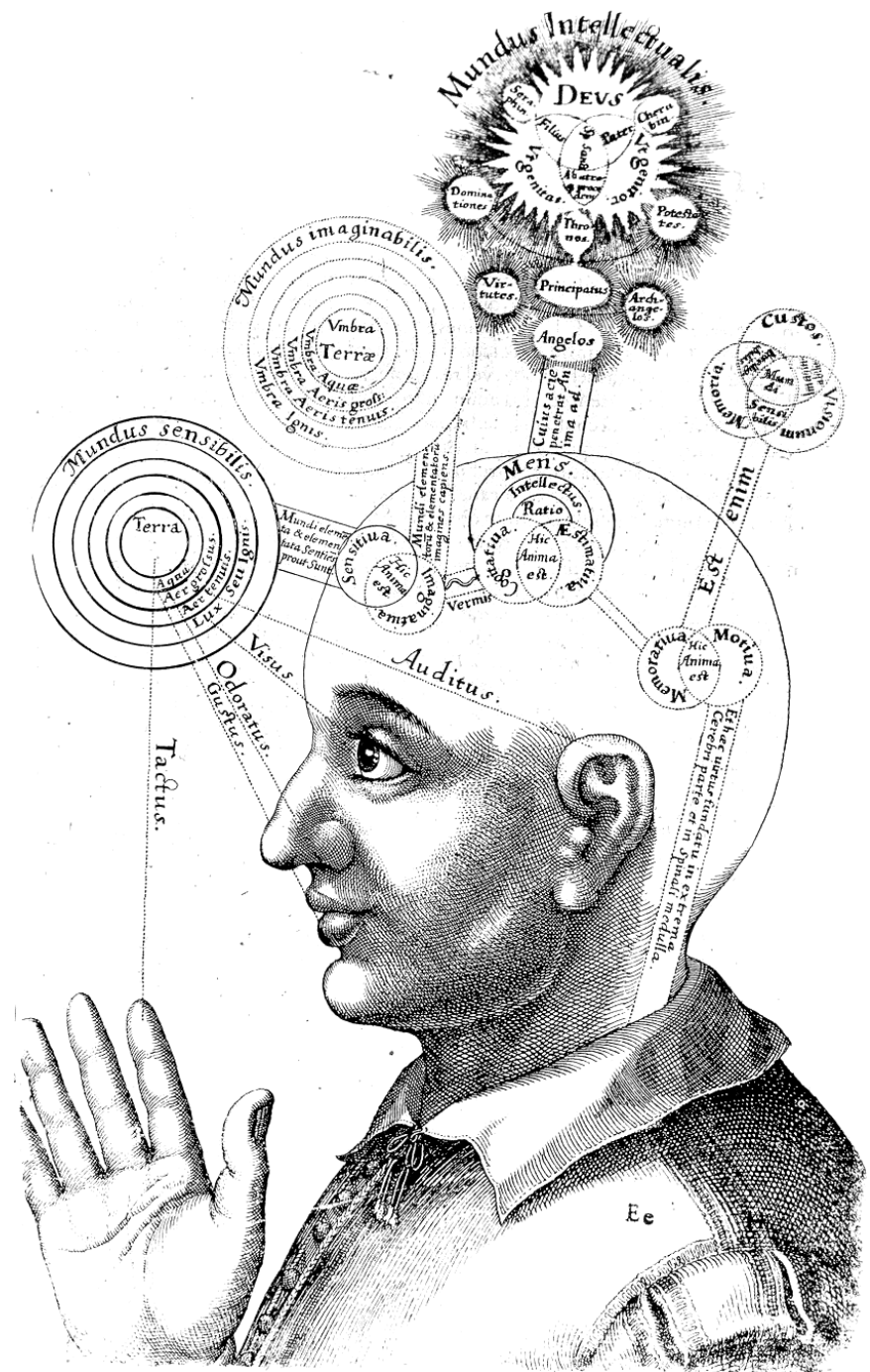
Sandini, G., A. Sciutti, and D. Vernon. Cognitive Robotics. In M. Ang, O. Khatib, and B. Siciliano (Eds.), Encyclopedia of Robotics. Springer, 2021.

What is Cognitive Robotics?

What is cognition?

Why is cognition useful in robotics?

How do cognitive robots work? ← Finally, we answer this question



Operation of a Cognitive Robot

Cognitive robots achieve their goals by

- **Perceiving** their environment
- Paying **attention** to the events that matter
- **Anticipating** the need for some action
- **Planning** what to do
- **Anticipating** the outcome as it executes the action
- **Learning** from the resultant interaction
- **Adapting** to change

Operation of a Cognitive Robot

Cognitive robots achieve their goals by

- **Perceiving** their environment

- Paying attention to the events that matter

- Anticipating

- Planning

- Anticipating the outcome as it executes the action

- Learning from the resultant interaction

- Adapting to change

Perception makes use of many sensory modalities, e.g. **vision**, **audition**, and **haptic** (tactile and kinesthetic)

(see <https://www.youtube.com/watch?v=HH6QD0MgqDQ>)

Operation of a Cognitive Robot

Cognitive robots achieve their goals by

- Perceiving their environment
- Paying **attention** to the events that matter

- Anticipating

Selective

selecting a given feature or object

- Planning

- Anticipating

Restrictive

restricting what to look for or where to look for it

- Learning

- Adapting

Suppressive

suppressing features, objects, or locations that are deemed to be not relevant

[Kotseruba and Tsotsos 2020]

Operation of a Cognitive Robot

Cognitive robots achieve their goals by

- Perceiving their environment
- Paying attention to the events that matter
- **Anticipating** the need for some action
- Planning
- Anticipating
- Learning
- Adapting

Anticipation; also referred to as **prospection**

The anticipated action is often associated with achieving a **goal**

Four modes of operation:

simulation, prediction, intention, and planning

[Szpunar et al. 2014]

Operation of a Cognitive Robot

Cognitive robots achieve their goals by

- Perceiving their environment
- Paying attention to the events that matter
- Anticipating the need for some action
- **Planning** what to do
- Anticipating the need for some action
- Learning from experience
- Adapting to new situations

Planning is sometimes effected by **reasoning** about the **current state of the world** or **anticipated futures states**

Exploits memories of past experience (**episodic memory**) and knowledge of the world (**semantic memory**)

Operation of a Cognitive Robot

Cognitive robots achieve their goals by

- Perceiving their environment
- Paying attention to the events that matter
- Anticipating the need for some action
- Planning what to do
- **Anticipating** the outcome as it executes the action
- Learning
- Adapting

Anticipating the outcome of a possible action can refer to the **actions of the robot** itself or the **actions of other agents** (people and other robots)

Operation of a Cognitive Robot

Cognitive robots achieve their goals by

- Perceiving
 - Learning from actions means that future actions can be more **effective** or more **efficient**
- Paying attention
- Anticipating
 - Often based on **reasoning**
- Planning
 - Sometimes referred to as **meta-cognition** or **meta-reasoning** (the focus is on improving the cognitive or reasoning process)
- Anticipating
- **Learning** from the resultant interaction
- Adapting to change

Operation of a Cognitive Robot

Cognitive robots achieve their goals by

- Perceiving their environment
- Paying attention to relevant information
- Anticipating future events
- Planning
- Anticipating the consequences of actions
- Learning
- **Adapting** to change

Adaptation is also achieved through learning

In this case the result of learning is a **new action policy** rather than an **improved action policy**

Operation of a Cognitive Robot

“Cognition is the process by which an autonomous system **perceives** its environment, **learns** from experience, **anticipates** the outcome of events, **acts** to pursue goals, and **adapts** to changing circumstances.”



Orchestrating all this requires a **cognitive architecture**

Operation of a Cognitive Robot

A key feature of cognitive robotics is the focus on **prospection** to augment immediate sensory-motor experience, both when **navigating** and **manipulating** objects in the robot's environment and when **interacting** with people.

Cognitive robots are able to carry out tasks effectively by **anticipating** the **effects of their own actions** as well as the **actions of the people around them**.

Being able to view the world from another person's perspective, a cognitive robot can **anticipate that person's intended actions** and **needs** and, consequently, it can interact safely while performing tasks in everyday situations.

This applies both during **direct interaction** (e.g. a robot assisting a customer in a supermarket) and **indirect interaction** (e.g. a robot stacking shelves while customers are shopping).

Operation of a Cognitive Robot

A Final Thought ...

There are **two** reasons people study cognitive robotics

1. They want to build smart robots
2. They want to understand cognition



Embodied Cognition



Summary

- Today we discussed

- 1- Introduction to cognitive robotics

- Next week

- 1- We will continue discussion other topics in cognitive robotics